

99.500 Applied Mathematics for Engineering

Homework 3 (Q1 is compulsory, followed by either Q2 or Q3)

March 24, 2026

1. A solid sphere, initially at a uniform temperature T_0 , is suddenly immersed in a constant-temperature bath maintained at T_∞ . It is assumed that the bath is not vigorously stirred, and the external heat-transfer resistance is negligible, implying that the surface temperature of the sphere remains at T_∞ .
 - (a) Determine the unsteady-state temperature profile within the sphere.
 - (b) Derive an expression for the sphere's average temperature as a function of time.
2. Use the separation of variables to find the solution $C(x, t)$ for the following problem:

$$\frac{\partial C}{\partial t} = \frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial C}{\partial x} \right) \quad (1)$$

$$C(x, 0) = 1 \quad (2)$$

$$\frac{\partial C}{\partial x} \Big|_{x=1} = 0 \quad (3)$$

$$\alpha \frac{\partial C}{\partial x} \Big|_{x=\beta} = C(\beta, t) \quad (4)$$

where α and β are constants.

3. Use the separation of variables to find the solution $C(x, t)$ for the following problem:

$$\frac{\partial C}{\partial t} = \frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial C}{\partial x} \right) - \delta C \quad (5)$$

$$C(x, 0) = 1 \quad (6)$$

$$\frac{\partial C}{\partial x} \Big|_{x=0} = 0 \quad (7)$$

$$C(1, t) = 0 \quad (8)$$

where δ is constant.